

Mobile Long-Term Traffic Flow Forecasting for Low-Altitude Traffic Surveillance

This is the appendix to the paper entitled ‘Mobile Long-Term Traffic Flow Forecasting for Low-Altitude Traffic Surveillance’, submitted to IEEE Transactions on Intelligent Transportation Systems.

APPENDIX A ADDITIONAL QUANTITATIVE RESULTS

We add comparisons with SE-LLM [1], FACT [2], and PhaseFormer [3], all accepted at ICLR 2026. Traffic dataset is adopted since it is the only dataset shared by all baselines. Under the same data splits, input-prediction settings, normalization, and metrics, MFNet continues to obtain the best average performance among the evaluated methods, as reported in Fig. A.1.

APPENDIX B LONG-TERM TRAFFIC FLOW FORECASTING WITH MISSING VALUES

To evaluate the robustness of the proposed MFNet under imperfect data conditions, we conduct additional experiments on long-term traffic flow (LTF) forecasting with missing

values. Specifically, missing values are randomly injected into the input sequences with missing ratios of {5%, 10%, 15%, 20%}, simulating realistic scenarios such as sensor failures, occlusions, and communication losses in intelligent transportation systems.

The evaluation is performed using the Cucconi rank test [4], which measures the similarity between the distributions of the predicted sequences and the corresponding ground truths. The Cucconi test is a nonparametric statistical method that jointly evaluates differences in both location and scale, without requiring prior assumptions on the underlying data distribution. This property makes it particularly suitable for traffic flow data, which are typically non-Gaussian and exhibit heteroscedastic characteristics. Its format is as follows.

Given two time series $\mathbf{X}_1 = \{x_{11}, \dots, x_{1n_1}\}$ and $\mathbf{X}_2 = \{x_{21}, \dots, x_{2n_2}\}$ ($n = n_1 + n_2$), the Cucconi test statistic (C), can be computed as follows:

$$\begin{aligned} C &= \frac{u^2 + v^2 - 2\rho uv}{2(1 - \rho^2)} < -\ln \alpha \\ \rho &= \frac{2(n^2 - 4)}{(2n + 1)(8n + 11)} - 1 \\ u &= \frac{6 \sum_{j=1}^{n_1} W_{1j}^2 - n_3}{n_4} \\ v &= \frac{6 \sum_{j=1}^{n_1} (n + 1 - W_{1j})^2 - n_3}{n_4} \end{aligned} \quad (\text{B.1})$$

where $n_3 = n_1(n + 1)(2n + 1)$ and $n_4 = \sqrt{n_1 n_2 (n + 1)(2n + 1)(8n + 11)/5}$. u is based on the standardized sum of the squared ranks W_{1j} , the rank of element x_{1j} in the pooled sample $\mathbf{X} = \mathbf{X}_1, \mathbf{X}_2 = x_{11}, \dots, x_{1n_1}, x_{21}, \dots, x_{2n_2}$, and v is based on the standardized sum of the squared contrary ranks $(n + 1 - W_{1j})^2$. ρ is the correlation coefficient between u and v . Under the null hypothesis, the observed value of the two-sample C statistic is expected to be close to 0, as shown in Fig. B.1.

The compared methods include TimeXer, which is the only baseline among the selected models that is specifically designed to handle missing values, as well as two recent

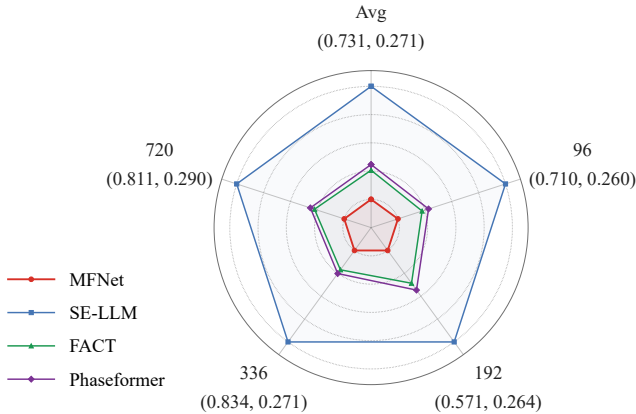


Fig. A.1. Traffic-domain MAE comparison across prediction lengths 96, 192, 336, and 720, together with their average. Lower MAE indicates better performance.

TABLE B.1
CUCCONI TEST STATISTICS OF THE FOUR MODELS UNDER DIFFERENT MISSING RATIOS ON FOUR DATASETS (PREDICTION SEQUENCE LENGTH = 720)

Models	PeMSD3				PeMSD4				PeMSD7				PeMSD8			
	5%	10%	15%	20%	5%	10%	15%	20%	5%	10%	15%	20%	5%	10%	15%	20%
MFNet	1.47	1.58	1.89	1.91	1.32	1.42	1.67	2.18	1.02	1.04	1.24	1.48	1.15	1.21	1.48	1.71
TimeXer	<u>1.99</u>	<u>2.13</u>	2.97	<u>3.27</u>	<u>1.73</u>	<u>1.90</u>	<u>2.64</u>	4.04	1.84	<u>1.84</u>	2.60	3.03	<u>1.52</u>	<u>1.58</u>	<u>2.34</u>	4.47
Time-VLM	2.74	3.07	3.63	4.93	2.53	2.58	3.51	4.62	<u>1.68</u>	2.03	<u>2.30</u>	<u>2.98</u>	1.88	2.17	3.64	3.53
FSTLLM	2.36	2.49	<u>2.79</u>	3.63	2.08	2.19	4.23	<u>3.33</u>	1.76	2.14	2.83	3.74	1.91	2.15	3.41	<u>2.66</u>

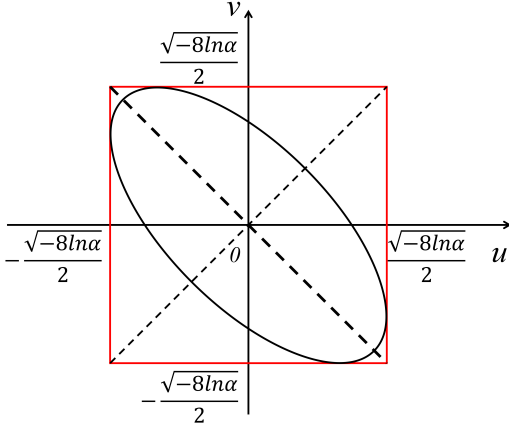


Fig. B.1. The acceptable realm of the null hypothesis of Cucconi test. The pair (u, v) is inside the ellipse if the null hypothesis is accepted otherwise the null hypothesis is rejected.

large language model (LLM)-based approaches, namely Time-VLM [5] and FSTLLM [6], which have demonstrated strong performance in long-term forecasting tasks. The significance level α is set to 0.05, corresponding to $-\ln \alpha \approx 3$.

The experimental results in Table B.1 show that MFNet consistently achieves Cucconi statistics that are significant under the null hypothesis across all missing ratios, indicating that the predicted distributions are statistically indistinguishable from the ground truth distributions. Moreover, MFNet obtains the lowest Cucconi statistics among all compared methods, demonstrating the highest degree of distributional alignment and robustness. These results verify that MFNet maintains stable performance under missing data conditions and is well-suited for real-world intelligent transportation system (ITS) applications where data incompleteness is unavoidable.

APPENDIX C GRADIENT STABILITY ANALYSIS

Due to the operations in the frequency domain, the gradient stability of the CTMLP module needs to be validated. Therefore, the average gradients of the layers belonging to the first CTMLP module during the first epoch of training process are visualized in Fig. C.1. As shown in Fig. C.1, the gradient

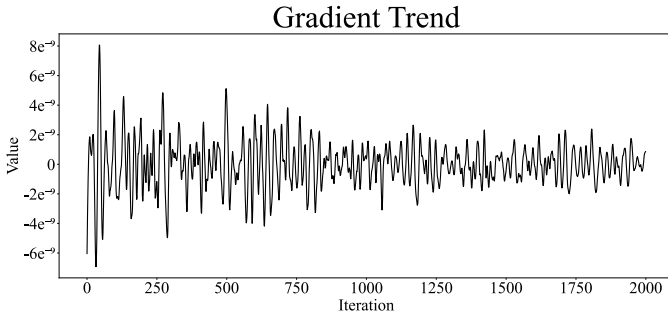


Fig. C.1. The average gradients of the layers belonging to the first CTMLP module during the first epoch of training process. The dataset is PeMSD8 and the prediction length is 720. This epoch contains nearly 2000 iterations and the gradient trend gradually converges with the growth of iterations.

magnitudes remain within a reasonable range throughout the training process and converge with the growth of iterations, indicating that neither vanishing nor exploding gradients occur. This suggests that CTMLP maintains stable gradient flow, likely due to the use of residual connections and the GELU activations with appropriate Kaiming initialization.

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